

Public-Data Octave-Band Noise Assessment Calculator

Project Design and Reviewer Feedback Brief

Purpose	Build and validate a reusable Excel workbook that combines published octave-band sound-power data, applies A-weighting, compares operating scenarios, and communicates dominant sources and bands.
Initial case	A mechanical-equipment group using compatible public data for a CRAC condenser, VRF condensers, and a toilet extract fan.
Current status	Design only. The workbook, calculations, and results have not yet been completed.
Intended use	Portfolio planning, structured learning based on professional advice, and feedback from a friend or practitioner familiar with acoustics, HVAC, engineering, or technical analysis.

Design boundary

Version 1 combines source sound-power spectra and operating quantities. It does not claim to predict sound pressure at a receiver, satisfy a noise criterion, or replace a standards-based environmental noise model.

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Technical Audio Engineering Development Portfolio

1. Project Rationale and Case Definition

This project responds to professional guidance from acoustical consultant Ray Plasse emphasizing practical Excel ability, octave-band sound levels, A-weighting, decibel summation, and the distinction between sound power and sound pressure. The aim is to convert those fundamentals into a small but audible engineering tool rather than a one-formula worksheet.

Proposed question

How do multiple items of mechanical equipment combine spectrally, which equipment and octave bands dominate the total sound power, and how do credible operating changes affect the combined result?

Public source dataset

The initial spectra will be transcribed from an Environment Agency-hosted 2024 noise assessment prepared by Ove Arup & Partners. Its tables report compatible octave-band sound-power levels for multiple mechanical equipment categories. The workbook will preserve the source URL, report title, table identifier, units, and transcription notes.

Equipment	63	125	250	500	1k	2k	4k	8k	LwA*
CRAC condenser	94	94	83	74	71	67	59	54	92
VRF condenser	82	72	74	72	67	63	57	53	73
Toilet extract fan	63	68	69	73	70	69	62	54	75

* LwA is the report's stated A-weighted sound-power total. The calculator will independently recompute A-weighted totals from the octave-band data as a transcription and formula check.

Initial operating scenarios

Scenario	CRAC	VRF	Extract	Reason
Normal	1	2	1	Baseline equipment group
Reduced	1	1	1	One VRF inactive
Ventilation only	0	0	1	Isolate extract contribution
Doubled system	2	4	2	Known +3.01 dB validation case

2. Proposed Workbook Architecture

The workbook will separate user inputs, reference data, calculations, outputs, and validation so another person can audit each result without reverse-engineering hidden formulas.

Sheet 1 - Read Me

- Purpose, scope, definitions, editable-cell convention, and limitations.
- Clear distinction between sound power (source property) and sound pressure (receiver/environment dependent).
- Version status and a statement that the workbook is educational portfolio work, not a professional assessment.

Sheet 2 - Published Source Data

- One row per equipment source with octave-band Lw values from 63 Hz through 8 kHz.
- Source document, table/page, URL, reported overall LwA, transcription checker, and notes.
- A visible A-weighting reference table; no unexplained corrections embedded inside formulas.

Sheet 3 - Scenario Inputs

- Scenario name, equipment count, enabled/disabled status, and optional user-entered per-band attenuation.
- Baseline values preloaded from the public source table; users change quantities rather than retyping spectra.
- Input validation preventing negative quantities and warning when incompatible units are introduced.

Sheet 4 - Calculations

- Quantity adjustment for identical independent sources.
- Energy-domain combination of all active sources within every octave band.
- Overall unweighted and A-weighted totals, source contributions, and scenario deltas.
- Formula cells remain visible and protected from accidental input edits where practical.

Sheet 5 - Results Dashboard

- Octave-band spectrum chart comparing baseline and selected alternative.
- Overall Lw and LwA by scenario.
- Dominant source, dominant octave band, and the size of each proposed change.
- A short conclusion field prompting an engineering interpretation instead of presenting numbers alone.

Sheet 6 - Validation

- Known-result checks, independent hand calculations, formula-error scan, and source-data reconciliation.
- Pass/fail indicators that remain visible in the portfolio version.

3. Calculation and Validation Plan

Core equations

Purpose	Planned expression	Interpretation
Identical units	$L_{total} = L_{single} + 10 \log_{10}(N)$	Accounts for N independent identical sources.
Combine levels	$L_{total} = 10 \log_{10}(\sum(10^{(L_i/10)}))$	Adds acoustic energy, not dB values arithmetically.
A-weighted total	$10 \log_{10}(\sum(10^{((L_i + A_i)/10)}))$	Applies the published A correction A_i before log summation.
Scenario change	$\Delta L = L_{scenario} - L_{baseline}$	Reports the effect of operating or mitigation changes.

Required validation cases

- One enabled source at quantity 1 reproduces its published octave-band spectrum exactly.
- Two identical sources produce +3.0103 dB in every band and in the overall total.
- Ten identical sources produce +10.0000 dB.
- A disabled or zero-quantity source contributes no energy and produces no spreadsheet error.
- Recomputed A-weighted totals are compared with the report's stated totals; differences are documented rather than silently forced to match.
- At least one octave band and one overall result are independently calculated outside the main formula chain.

Version 1 exclusions

Receiver sound-pressure prediction, distance attenuation, barriers, ground effects, atmospheric absorption, reflections, tonality penalties, background corrections, NC/NR/RC rating, and regulatory compliance are intentionally excluded. Those require a defined application, appropriate standards, and additional inputs.

Why exclude receiver prediction initially?

Published source sound power can be combined consistently. Converting that result into sound pressure requires a propagation path or room transfer function. Trane's technical guidance explicitly distinguishes sound power from sound pressure and notes that occupied-space criteria require application-specific transfer information. The first version therefore emphasizes correct source arithmetic and transparent limitations.

4. Development Phases and Evidence

Phase	Work	Completion evidence
1. Source audit	Verify units, bands, tables, and transcription from the public report.	Source log and reconciliation checks
2. Core calculator	Build quantity, octave-band combination, A-weighting, and overall total tables.	Validated workbook with no formula errors
3. Scenarios	Implement normal, reduced, isolated, and doubled-system cases.	Scenario table and comparison chart
4. Validation	Run known-result and independent checks; document discrepancies.	Visible test sheet with pass/fail results
5. Interpretation	Identify dominant source/band and explain scenario changes.	One-page technical summary
6. External review	Send this brief and/or workbook to a knowledgeable reviewer.	Feedback log and justified revisions

Definition of done for the first portfolio release

- Published source data are traceable and unit-compatible.
- All derived outputs use visible formulas.
- Known validation cases pass and any source-total discrepancy is explained.
- The dashboard communicates both spectral and overall results.
- The write-up distinguishes completed behavior, assumptions, limitations, and future work.
- External feedback is summarized as review input, not presented as endorsement or formal supervision.

Potential later extensions

- A standards-informed outdoor propagation module using a defined receiver geometry.
- Manufacturer-specific equipment comparisons under matched operating conditions.
- Import of measured octave-band spectra after calibrated equipment becomes available.
- A separate vehicle-cabin tool only after the measurement and transfer-function requirements are understood.

Portfolio claim

Designed and validated a reusable octave-band source-noise assessment workbook using publicly reported mechanical-equipment sound-power spectra, with scenario analysis, A-weighted totals, contribution analysis, and explicit technical limitations.

The claim above is prospective and must not be used until the workbook and validation are complete.

5. Reviewer Feedback Request

This page is written so the document can be sent directly to a friend or practitioner. The request is for practical feedback on scope and relevance, not approval of work that has not yet been completed.

Suggested note to accompany this PDF

I am developing this as an introductory acoustical-analysis portfolio project based on advice to strengthen Excel, octave-band, A-weighting, and decibel-summation skills. Would you be willing to look at the proposed design and tell me whether it resembles a useful entry-level exercise, where it connects to real work, and what I should change before building it? I am especially interested in whether the Version 1 boundary is appropriate and what a credible next stage would be.

Questions for the reviewer

1. Does combining public sound-power spectra and operating quantities resemble a useful task for an entry-level acoustical consultant, or is the exercise missing an essential practical step?
2. Is keeping receiver sound-pressure prediction out of Version 1 a sound boundary, or should a simple propagation case be included from the beginning?
3. Are the proposed source types and operating scenarios realistic enough to demonstrate useful judgment?
4. Which workbook outputs would matter most in actual work: overall dBA, octave-band dominance, source contribution, scenario delta, or another result?
5. What mistakes do beginners commonly make when working with published equipment sound data?
6. Which standard, calculation method, or software workflow would be the most logical next step after this calculator works?
7. Would this project be useful evidence for acoustical consulting, HVAC noise-control, system evaluation, or another discipline? How should that relationship be described without overstating it?

Reference sources

1. Ove Arup & Partners Ltd., *Noise Impact Assessment*, Rev. 1, 6 June 2024, equipment sound-power tables on report pages 9-10. Environment Agency consultation host:
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2. Trane, *Evaluating Sound Data*, Engineers Newsletter Live course material, 2015, especially pages 24-30 on A-weighting, octave-band summation, and sound power versus sound pressure:
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3. Daikin Applied, *HVAC Acoustic Fundamentals*, Application Guide AG 31-010, including octave-band path summation and sound-power data examples:
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